

Sequences — Solutions

Problem 1. (Nigeria Senior MO 2nd Round 2022 Problem 4) Define sequence $(a_n)_{n=1}^{\infty}$ by $a_1 = a_2 = a_3 = 1$ and $a_{n+3} = a_{n+1} + a_n$ for all $n \geq 1$. Also, define sequence $(b_n)_{n=1}^{\infty}$ by $b_1 = b_2 = b_3 = b_4 = b_5 = 1$ and $b_{n+5} = b_{n+4} + b_n$ for all $n \geq 1$. Prove that there exists $N \in \mathbb{N}$ such that $a_n = b_{n+1} + b_{n-8}$ for all $n \geq N$.

Solution. We extend the sequence (b_n) by setting $b_0 = b_{-1} = b_{-2} = b_{-3} = 0$, $b_{-4} = 1$ and $b_{-5} = 0$. Then the relation $b_{n+5} = b_{n+4} + b_n$ holds for all $n \geq -5$. We prove by induction that $a_n = b_{n+1} + b_{n-8}$ for any $n \geq 3$.

We first check the base cases $n = 3, 4, 5$ as follows.

$$\begin{aligned} a_3 &= 1 = 1 + 0 = b_4 + b_{-5} \\ a_4 &= a_2 + a_1 = 1 + 1 = b_5 + b_{-4} \\ a_5 &= a_3 + a_2 = 1 + 1 = 2 + 0 = b_6 + b_{-3} \end{aligned}$$

For the inductive step, we assume $a_{n-3} = b_{n-2} + b_{n-11}$ and $a_{n-2} = b_{n-1} + b_{n-10}$ for some $n \geq 6$. Then we have $a_n = a_{n-2} + a_{n-3} = (b_{n-1} + b_{n-10}) + (b_{n-2} + b_{n-11})$. We need to check that $b_{n+1} + b_{n-8} = b_{n-1} + b_{n-2} + b_{n-10} + b_{n-11}$. Indeed,

$$\begin{aligned} & (b_{n+1} + b_{n-8}) - (b_{n-1} + b_{n-2} + b_{n-10} + b_{n-11}) \\ &= (b_n + b_{n-4}) - b_{n-1} - b_{n-2} + b_{n-8} - b_{n-10} - b_{n-11} \\ &= (b_{n-1} + b_{n-5}) - b_{n-1} - b_{n-2} + b_{n-4} + b_{n-8} - b_{n-10} - b_{n-11} \\ &= -(b_{n-3} + b_{n-7}) + b_{n-4} + b_{n-5} + b_{n-8} - b_{n-10} - b_{n-11} \\ &= -(b_{n-4} + b_{n-8}) + b_{n-4} + b_{n-5} - b_{n-7} + b_{n-8} - b_{n-10} - b_{n-11} \\ &= (b_{n-6} + b_{n-10}) - b_{n-7} - b_{n-10} - b_{n-11} \\ &= b_{n-6} - b_{n-7} - b_{n-11} = 0. \end{aligned}$$

This proves the inductive step, and so the result holds for any $n \geq 3$. □

Problem 2. (Bosnia and Herzegovina TST for Balkan MO 2022 Problem 1) Let $a_1, a_2, a_3, \dots$ be an infinite sequence of nonnegative real numbers such that for all positive integers k the following conditions hold:

(i) $a_k - 2a_{k+1} + a_{k+2} \geq 0$;

(ii) $\sum_{j=1}^k a_j \leq 1$.

Prove that for all positive integer k the inequality $0 \leq a_k - a_{k+1} < \frac{2}{k^2}$ holds.

Solution. Let $b_k = a_k - a_{k+1}$ for all $k \geq 1$. The first condition means $\{b_k\}$ is monotonic decreasing.

If $b_k < 0$ for some $k \geq 1$, then we have $b_j < 0$ for all $j \geq k$. This implies $a_j < a_{j+1}$ for all $j \geq k$. This easily yields $a_j \geq a_{k+1} > 0$ for all $j \geq k+1$. However, this implies

$$1 \geq \sum_{j=1}^{k+\ell} a_j \geq \sum_{j=k+1}^{k+\ell} a_j \geq \ell a_{k+1}$$

for any $\ell \geq 1$. This does not hold if $\ell > \frac{1}{a_{k+1}}$. Therefore, we must have $b_k \geq 0$.

If $b_k \geq \frac{2}{k^2}$ for some $k \geq 1$, then we have $b_j \geq \frac{2}{k^2}$ for all $j \leq k$. This implies

$$a_i = \sum_{j=i}^k b_j + a_{k+1} \geq (k+1-i) \cdot \frac{2}{k^2} + 0 = \frac{2(k+1-i)}{k^2}$$

for all $i \leq k$. However, this implies

$$1 \geq \sum_{i=1}^k a_i \geq \sum_{i=1}^k \frac{2(k+1-i)}{k^2} = \frac{k+1}{k},$$

contradiction. Therefore, we must have $b_k < \frac{2}{k^2}$. □

Problem 3. (Latvia TST for Baltic Way 2018 Problem 3) Let $a_1, a_2, \dots$ be an infinite sequence of integers that satisfies $a_{n+2} = a_{n+1} + a_n$ for all $n \geq 1$. There exists a positive integer k such that $a_k = a_{k+2018}$. Prove that there exists a term of the sequence which is equal to zero.

Solution. Let α and β be the roots of the characteristic equation $t^2 - t - 1 = 0$. Then $\alpha\beta = -1$ by Vieta's theorem. It is well-known that $a_n = A\alpha^n + B\beta^n$ for some constants A, B . Since $a_k = a_{k+2018}$, we have

$$A(\alpha^k - \alpha^{k+2018}) = B(\beta^{k+2018} - \beta^k).$$

Note that $\alpha^k - \alpha^{k+2018}$ and $\beta^{k+2018} - \beta^k$ are nonzero as $\alpha, \beta \neq 0, \pm 1$. Therefore, we can write $A = (\beta^{k+2018} - \beta^k)C$ and $B = (\alpha^k - \alpha^{k+2018})C$ for some constant C . It follows that

$$\begin{aligned} A\alpha^{k+1009} + B\beta^{k+1009} &= (\beta^{k+2018} - \beta^k)C\alpha^{k+1009} + (\alpha^k - \alpha^{k+2018})C\beta^{k+1009} \\ &= C\alpha^{k+1009}\beta^{k+1009}(\beta^{1009} - \beta^{-1009} + \alpha^{-1009} - \alpha^{1009}) \\ &= C\alpha^{k+1009}\beta^{k+1009}(\beta^{1009} - \alpha^{1009})\left(1 + \frac{1}{(\alpha\beta)^{1009}}\right) \\ &= C\alpha^{k+1009}\beta^{k+1009}(\beta^{1009} - \alpha^{1009})\left(1 + \frac{1}{(-1)^{1009}}\right) = 0. \end{aligned}$$

□

Problem 4. (Dutch TST 2018 Test 1 Problem 3) Let $n \geq 0$ be an integer. A sequence $a_0, a_1, a_2, \dots$ of integers is defined as follows: we have $a_0 = n$ and for $k \geq 1$, a_k is the smallest integer greater than a_{k-1} for which $a_k + a_{k-1}$ is the square of an integer. Prove that there are exactly $\lfloor \sqrt{2n} \rfloor$ positive integers that cannot be written in the form $a_k - a_\ell$ with $k > \ell \geq 0$.

Solution. Let $c = \lfloor \sqrt{2n} \rfloor$. Then $c^2 \leq 2n < (c+1)^2$. As $a_0 + a_1 > 2a_0 = 2n \geq c^2$, we have $a_1 = (c+1)^2 - n > a_0$. We shall prove by induction that

$$a_k + a_{k-1} = (c+k)^2 \quad (1)$$

for any $k \geq 1$. The base case $k = 1$ has been verified.

Consider the inductive step for the case k . Since $a_k + a_{k-1} > a_{k-2} + a_{k-1} = (c+k-1)^2$, it suffices to check $(c+k)^2 - a_{k-1} > a_{k-1}$ (so that $a_k = (c+k)^2 - a_{k-1}$), i.e. $a_{k-1} < \frac{(c+k)^2}{2}$. If $k = 2$, we need to prove

$$(c+1)^2 - n < \frac{(c+2)^2}{2}.$$

This is equivalent to $c^2 < 2n + 2$, which is true. If $k \geq 3$, we have $a_{k-2} + a_{k-3} = (c+k-2)^2$ and hence $a_{k-2} > \frac{(c+k-2)^2}{2}$. It follows that

$$a_{k-1} = (c+k-1)^2 - a_{k-2} < (c+k-1)^2 - \frac{(c+k-2)^2}{2} < \frac{(c+k)^2}{2}$$

by the QM-AM inequality. This completes the inductive step.

Using (1), we deduce

$$a_{k+2} - a_k = (a_{k+2} + a_{k+1}) - (a_{k+1} + a_k) = (c+k+2)^2 - (c+k+1)^2 = 2c + 2k + 3$$

and

$$(a_{k+3} - a_{k+2}) - (a_{k+1} - a_k) = (a_{k+3} - a_{k+1}) - (a_{k+2} - a_k) = (2c + 2(k+1) + 3) - (2c + 2k + 3) = 2$$

for any $k \geq 0$. This easily implies $a_{2k+1} - a_{2k} = a_1 - a_0 + 2k$ and $a_{2k+2} - a_{2k+1} = a_2 - a_1 + 2k$ for any $k \geq 0$. Therefore, every number $\geq a_1 - a_0$ and has the same parity as $a_1 - a_0$ is the difference of two consecutive terms, and hence can be represented as the desired form. Similarly, every number $\geq a_2 - a_1$ and has the same parity as $a_2 - a_1$ can be represented as the desired form. Note that $a_1 - a_0 = (c+1)^2 - 2n$ and $a_2 - a_1 = (c+2)^2 - 2a_1 = 2n - c^2 + 2$ have different parities.

On the other hand, all other numbers cannot be represented as the desired form. Indeed, if $m = a_k - a_\ell$ for some $k > \ell \geq 0$ where m is not of the above forms, then we must have $k \geq \ell + 2$. So $m > a_k - a_{k-1}$ and $m > a_{k-1} - a_{k-2}$, which easily implies $m > a_2 - a_1$ and $m > a_1 - a_0$. But then m must be of the above forms since it has the same parity as $a_2 - a_1$ or $a_1 - a_0$.

Therefore, it remains to count the number of positive integers which are not of the above forms. The number of positive integers $< a_1 - a_0$ having the same parity as $a_1 - a_0$ is $\left\lfloor \frac{a_1 - a_0 - 1}{2} \right\rfloor$. Similarly,

the number of positive integers $< a_2 - a_1$ having the same parity as $a_2 - a_1$ is $\left\lfloor \frac{a_2 - a_1 - 1}{2} \right\rfloor$. So the total number is

$$\left\lfloor \frac{a_1 - a_0 - 1}{2} \right\rfloor + \left\lfloor \frac{a_2 - a_1 - 1}{2} \right\rfloor = \left\lfloor \frac{c^2 + 2c - 2n}{2} \right\rfloor + \left\lfloor \frac{2n - c^2 + 1}{2} \right\rfloor = \left\lfloor \frac{c^2}{2} \right\rfloor + \left\lfloor \frac{-c^2 + 1}{2} \right\rfloor + c = c$$

as desired. $\square$

Problem 5. Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence of real numbers such that

$$a_{n+2} = |a_{n+1}| - a_n$$

for all $n \geq 1$. Prove that there exists $d \in \mathbb{N}$ such that $a_{n+d} = a_n$ for all $n \in \mathbb{N}$.

Solution. If $a_n, a_{n+1}, a_{n+2}, a_{n+3} > 0$ for some n , then $a_n + a_{n+2} = a_{n+1}$ and $a_{n+1} + a_{n+3} = a_{n+2}$. Summing these, we obtain $a_n + a_{n+3} = 0$, contradiction. So there are infinitely many nonpositive terms. Also, note that one of a_n, a_{n+2} must be nonnegative since $a_n + a_{n+2} = |a_{n+1}| \geq 0$.

If $a_{n+1}, a_{n+2} \leq 0$, let $a_{n+1} = -b$ and $a_{n+2} = -c$ for some $b, c \geq 0$. Then $a_{n+3} = b+c$, $a_{n+4} = b+2c$, $a_{n+5} = c$, $a_{n+6} = -b-c$, $a_{n+7} = b$, $a_{n+8} = 2b+c$, $a_{n+9} = b+c$, $a_{n+10} = -b$ and $a_{n+11} = -c$. By induction (forward and backward), we have $a_{m+9} = a_m$ for any $m \in \mathbb{N}$.

If no two consecutive terms are both nonpositive, suppose $a_{n+1} = -b$ and $a_{n+2} = c$ for some $b, c \geq 0$. Then $a_n = b-c \geq 0$ (otherwise there are two consecutive nonpositive terms). We have $a_{n+3} = b+c$, $a_{n+4} = b$, $a_{n+5} = -c$ and $a_{n+6} = c-b$. Note that both a_{n+5} and a_{n+6} are nonpositive, contradiction. So we are done. $\square$

Problem 6. (China TST 2008 Test 1 Problem 3) Suppose we colour every positive integer red or blue randomly. Prove that there exists an infinite sequence of positive integers $a_1 < a_2 < \dots$ such that the infinite sequence of positive integers $a_1, \frac{a_1 + a_2}{2}, a_2, \frac{a_2 + a_3}{2}, a_3, \dots$ has the same colour.

Solution. The result is trivial if there exists an infinite arithmetic progression whose every term has the same colour. In the following, we assume such an arithmetic progression does not exist.

WLOG assume 1 is red. We try to construct a sequence $\{b_n\}$ as follows: $b_1 = 1$, and for each $k \geq 2$, b_k is the smallest odd positive integer greater than b_{k-1} such that both b_k and $\frac{b_{k-1} + b_k}{2}$ are red. If this gives an infinite sequence $\{b_n\}$, we are done. Otherwise, suppose b is the largest term of this finite sequence. Then we know that for every m , one of $b+m$ and $b+2m$ must be blue.

Next, there exists an odd positive integer $c_1 > b$ which is blue. We construct the sequence $\{c_n\}$ analogous to the sequence $\{b_n\}$. In other words, c_k is the smallest odd positive integer greater than c_{k-1} such that both c_k and $\frac{c_{k-1} + c_k}{2}$ are blue. Again, we may assume there exists a largest term c of this sequence. Then for every m , one of $c+m$ and $c+2m$ must be red.

Now, consider any blue number $d > c$. Then $2d - c$ is red, and so $\frac{(2d - c) + b}{2} = d - \frac{c - b}{2}$ is blue.

Let $s = \frac{c - b}{2} > 0$. Then we have proven that whenever $d > c$ is blue, so is $d - s$. Equivalently, whenever $d > c$ is red, so is $d + s$. As there must be some red number greater than c , this gives an infinite arithmetic sequence which is red, contradiction.

So such an infinite sequence must exist. $\square$

Problem 7. (Simon Marais Mathematics Competition 2017 Problem A2) Let $a_1, a_2, a_3, \dots$ be the sequence of real numbers defined by $a_1 = 1$ and

$$a_m = \frac{1}{a_1^2 + a_2^2 + \dots + a_{m-1}^2} \quad \text{for } m \geq 2.$$

Determine whether there exists a positive integer N such that

$$a_1 + a_2 + \dots + a_N > 2017^{2017}.$$

Solution. Yes. It suffices to show that $s_n = a_1 + a_2 + \dots + a_n$ diverges to positive infinity. Firstly, note that $a_m > 0$ and $a_m < \frac{1}{a_1^2} = 1$ for all $m \geq 2$. This implies

$$s_n = a_1 + a_2 + \dots + a_n \geq a_1^2 + a_2^2 + \dots + a_n^2 = \frac{1}{a_{n+1}}$$

for all $n \in \mathbb{N}$. If s_n does not diverge to positive infinity, it must converge to some $L \in \mathbb{R}$ since it is monotonic increasing. This implies $\{a_n\}$ converges to 0. However, this contradicts $s_n \geq \frac{1}{a_{n+1}}$ since $s_n \leq L$ but $\frac{1}{a_{n+1}}$ diverges to positive infinity. Therefore, s_n must diverge to positive infinity. $\square$

Problem 8. (Romania TST 2021 Day 3 Problem 3) Let α be a real number in the interval $(0, 1)$. Prove that there exists a sequence $(\varepsilon_n)_{n \geq 1}$ where each term is either 0 or 1 such that the sequence $(s_n)_{n \geq 1}$ defined by

$$s_n = \frac{\varepsilon_1}{n(n+1)} + \frac{\varepsilon_2}{(n+1)(n+2)} + \dots + \frac{\varepsilon_n}{(2n-1)2n}$$

satisfies the inequality

$$0 \leq \alpha - 2ns_n \leq \frac{2}{n+1}$$

for any $n \geq 2$.

Solution. We define ε_n such that

$$\sum_{j=1}^n \varepsilon_j = \lfloor \alpha n \rfloor$$

for all $n \in \mathbb{N}$. Note that each ε_n is 0 or 1 since

$$0 \leq \varepsilon_n = \lfloor \alpha n \rfloor - \lfloor \alpha(n-1) \rfloor < \alpha n - (\alpha(n-1) - 1) = \alpha + 1 < 2.$$

We first prove the following.

Claim. If $\sum_{j=1}^n a_j \geq \sum_{j=1}^n b_j$ for all $n \in \mathbb{N}$, then $\sum_{j=1}^n \frac{a_j}{(n+j-1)(n+j)} \geq \sum_{j=1}^n \frac{b_j}{(n+j-1)(n+j)}$.

Proof. Let $A_n = \sum_{j=1}^n a_j$ and $B_n = \sum_{j=1}^n b_j$. Note that $\{c_n\}$ defined by $c_j = \frac{1}{(n+j-1)(n+j)}$ is a decreasing sequence. Therefore, by Abel's summation formula, we have

$$\sum_{j=1}^n a_j c_j = c_n A_n + \sum_{i=1}^{n-1} (c_i - c_{i+1}) A_i \geq c_n B_n + \sum_{i=1}^{n-1} (c_i - c_{i+1}) B_i = \sum_{j=1}^n b_j c_j.$$

□

Now, since $\sum_{j=1}^n \varepsilon_j = \lfloor \alpha n \rfloor \leq \alpha n = \sum_{j=1}^n \alpha$ for all n , by the claim, we have

$$s_n = \sum_{j=1}^n \frac{\varepsilon_j}{(n+j-1)(n+j)} \leq \sum_{j=1}^n \frac{\alpha}{(n+j-1)(n+j)} = \alpha \sum_{j=1}^n \left(\frac{1}{n+j-1} - \frac{1}{n+j} \right) = \frac{\alpha}{2n}.$$

This implies $\alpha - 2ns_n \geq 0$.

Similarly, since $(\varepsilon_1 + 1) + \sum_{j=2}^n \varepsilon_j = \lfloor \alpha n \rfloor + 1 > \alpha n = \sum_{j=1}^n \alpha$ for all n , we have

$$s_n + \frac{1}{n(n+1)} = \frac{\varepsilon_1 + 1}{n(n+1)} + \sum_{j=2}^n \frac{\varepsilon_j}{(n+j-1)(n+j)} \geq \sum_{j=1}^n \frac{\alpha}{(n+j-1)(n+j)} = \frac{\alpha}{2n}.$$

This implies $\alpha - 2ns_n \leq \frac{2}{n+1}$. This completes the proof. □